

MICRO EAZY · TAILNET OPERATIONS · 11 SEPTEMBER 2026

Making servicesuite- in the main host

Yes for SQL and the Interchange. No for the bureau. One of the two relays can move to `servicesuite-in` tonight; the other physically cannot, and the reason is the same reason the whole estate exists. The IIS port conflict has a measured cause and a one-line fix that touches nothing on the shared box.

Tailnet `tail10c441.ts.net` · 8 nodes Measured from `pc`, 100.101.186.43
Standing rule never disrupt Micromart's shared server

STATE What is actually true right now

Every row below was probed from this workstation while writing this, not read off a config. Three things in it are not what the documentation says, and one of them is the thing you are about to fix.

FUNNEL ENDPOINTS · MEASURED 11 SEP 2026

ENDPOINT	ANSWERS	READING
servicesuite:443	LIVE	sql-relay · 1,021 served, 12 refused, writes:false. This is the relay production is using — not the one the deployment notes name.
salesmaster:443	LIVE	crb-relay · 0 served since 4 Sep. Metropolis's registered egress. Cannot move.
salesmaster:8443	STALE	A second sql-relay, 0 served, running an older build than either checkout — its /health has no writes field at all. Nothing points at it.
servicesuite-in:443	502	Funnel is on and correct; nothing is listening behind it . This is the node you want, already half-configured.
lms:443	NO FUNNEL	Already turned off. Nothing to do.

SERVICESUITE-IN · 100.115.241.36 · PORT BY PORT

PORT	STATE	OWNER
80	open · HTTP 200	IIS, serving.
443	open · TLS fails on IP	tailscaled — presents CN=servicesuite-in.tail10c441.ts.net issued by Let's Encrypt, proxying to a dead backend.
4420	open	SQL Server on this box.
2020	—	PostgreSQL. FastDialCallBoxMaster and serviceconnect, per CallBox\appsettings.json.
8260	—	FastDial PBX report endpoint.
8787 / 8788	closed	No relay running. Confirms the 502 above.

SERVICESUITE-IN IS 213.148.17.228 , AND IT IS NOT A SPARE BOX

CallBox\appsettings.json identifies it: CallBox and FastDial run here, against PostgreSQL on 213.148.17.228:2020 , with a PBX on 8260 and IIS on 80. **The standing rule about Micromart's shared server applies to this box too.** Everything in this runbook is additive — a new service on 8787 and one funnel mapping — and nothing in it stops or reconfigures an existing site. Keep it that way: no iisreset , and check what shares an app pool before touching one.

Two things worth your own eyes, since they are read off a config rather than measured: that Postgres connection names the **public** address with the postgres superuser and a weak password. If 2020 answers from outside your network, that is a bigger exposure than anything else on this page — confirm it, and if it is open, bind it to the tailnet and close the public port. It is the same argument as the SQL relay: the database should be reachable over the overlay, not the internet.

THE IIS ERROR, DIAGNOSED

That second row *is* your bug. Tailscale holds 100.115.241.36:443 . IIS's Default Web Site binds **All Unassigned** on 443, which makes HTTP.sys claim the wildcard 0.0.0.0:443 — and a wildcard claim and a specific-address claim on the same port cannot coexist. tailscaled starts early, wins the port, and the site you restart afterwards fails with *“the process cannot access the file because it is being used by another process”* (0x80070020). Nothing is corrupt and no reboot will help, because the winner is decided by start order.

DECISION Why “one funnel” is not reachable, and two is the right number

You asked to consolidate onto `servicesuite-in` and run a single funnel. The SQL half of that is right and should happen tonight. The bureau half cannot happen at all, and it is worth being precise about why, because the failure mode if it is attempted is invisible.

SQL relay MOVE IT

needs: a tailnet route to the lender databases

Micromart's SQL at `100.72.35.56,4230` is CGNAT space with no internet route. Any node on the tailnet can reach it. `servicesuite-in` qualifies, stays awake, and already has Funnel approved. **Move it here.**

CRB relay LEAVE IT

needs: a public egress IP on Metropolis's whitelist

Metropol answers registered *source* addresses. `salesmaster` egresses `102.214.69.233` over its OpenVPN TAP adapter — on the 2026-08-27 form, verified 8/8 ENTITLED. **It stays there.**

THE ASYMMETRY THAT MAKES THIS NON-NEGOTIABLE

A funnel changes where a relay can be **reached**. It does not change where the relay **calls from**. Funnelling the CRB relay on `servicesuite-in` would publish a perfectly healthy endpoint whose `/health` answers `{"ok":true}` — and every bureau call it made would be dropped at Metropolis's edge with no response, no `api_code` and nothing in any log. It looks exactly like the bureau being down. That is the single most expensive mistake available in this estate.

So the steady state is **two funnels on two nodes**, and both of them are on port 443 of their own node, which means no port suffix and no split: `servicesuite-in` publishes the SQL relay and the Interchange broker; `salesmaster` publishes the CRB relay and nothing else. Moving SQL off `salesmaster` actually *removes* the 443/8443 split that has been the source of the `SERVICESUITE_RELAY_URL` confusion — a URL missing its `:8443` silently reached the CRB relay, whose health check also says OK.

443 Making IIS and Tailscale coexist

Two fixes work. One of them touches a shared production IIS box during the week of a board demo, and one is a single reversible command. Take the second.

Recommended — move the funnel, leave IIS alone

Funnel accepts only 443, 8443 and 10000. Give 443 back to IIS and publish the relay on 8443. The URL carries a port; no human ever sees it — it lives in `SERVICESUITE_RELAY_URL` on Vercel and nowhere else.

```
# on servicesuite-in, as Administrator
tailscale funnel status          # see what is mapped before changing it
tailscale funnel --https=443 off # release 443 back to HTTP.sys
tailscale serve status          # confirm nothing still holds 443

# now IIS can start
Start-Website -Name "Default Web Site"

# republish the relay one port up
tailscale funnel --bg --https=8443 8787
```

CONFIRM THE PORT ACTUALLY CHANGED HANDS

```
# PID 4 = System = HTTP.sys (IIS). A tailscaled PID = Tailscale still holds it.
netstat -ano | findstr ":443"
tasklist /fi "pid eq 4"
netsh http show servicestate view=requestq | findstr /i "443"
```

The other fix, and why not to reach for it

You *can* keep Funnel on 443 by changing every IIS site on the box from **All Unassigned** to an explicit IP binding, so HTTP.sys claims only that address and leaves the `100.x` one free.

- It only works if **every** site on the box is changed — one sibling left on `*:443` re-registers the wildcard and the conflict returns.
- `servicesuite-in` and `salesmaster` host other people's production sites. A binding change there is a change to systems that are not ours, for a cosmetic gain.
- It buys you a URL with no `:8443`. Nothing reads that URL but an environment variable.

If you ever do need 443 on the tailnet name — a human-facing console, say — that is the moment to make the binding change deliberately, at an agreed time, having first listed what shares the box.

NEVER TYPE A BARE `TAILSCALE FUNNEL OFF` ON THESE NODES

Funnel state is **per node** and the bare form clears every mapping on whichever box you are logged into. On `salesmaster` that silently kills the bureau link. On `servicesuite` it takes the live SQL path down and all six systems go amber at once. Always name the port: `--https=443 off`.

RUN The cutover, in order

Each step depends on the one before it. The ordering principle throughout: stand the new thing up and prove it, *then* turn the old thing off. Nothing here is reached by the live site until step 6.

01

Give 443 back to IIS SERVICESUITE-IN

The commands above. Confirm the Default Web Site starts before going further — if it does not, stop here, because the diagnosis was wrong and everything downstream assumes it was right.

02

Put the relay bundle on the box SERVICESUITE-IN

Zip `relay-host\` from this repo and unzip to `C:\relay`. Keep the folder structure exactly — both relays resolve `.env` from the bundle root, and `sql-relay.ts` needs the `src\` tree beside it.

If the box has no route to the npm registry, use the offline bundle instead; it ships `node_modules` prebuilt.

```
powershell -ExecutionPolicy Bypass -File check-host.ps1
```

Install only what that script approves. It answers the which-box question with evidence rather than assumption.

03

Carry the secrets over unchanged SERVICESUITE-IN

Copy `SERVICESUITE_RELAY_SECRET` and the connection strings from the `servicesuite` node's `.env` byte for byte. Reusing the existing secret means **Vercel needs no redeploy for the secret** — only the URL changes, and that is one variable.

Set the write switch to exactly the one grant:

```
SQL_RELAY_ALLOW_PROCS=Notifications.dbo.sp_InsertsmsAndEmails  
SQL_RELAY_ALLOW_WRITES=false
```

`SQL_RELAY_ALLOW_WRITES` stays false. The allowlist is what opens the outbox now, and it opens nothing else.

04

Start it and read the boot banner

SERVICESUITE-IN

The banner is the verification. It prints the grants it just opened, by name, with their orgs and their permitted parameters — so whoever restarts the relay sees which doors they opened without reading the `.env` back.

```
SQL relay listening on http://127.0.0.1:8787
writes:  refused (read-only)
allowed: 1 write grant(s) even while read-only
        Notifications.dbo.sp_InsertsmsAndEmails
        orgs:  micromart, micromart-fintech
        params: receiver, bodyMessage, companyid

warming connection pools...
✓ Micromart Africa pool open (2340ms)
✓ Axe Capital pool open (1228ms)
```

Both pools must open. If Axe does not answer, the tailnet route to `100.103.154.73,4420` is the problem, not the relay. First start sits 8–20 seconds warming — that is deliberate, it absorbs the cold TDS handshake at boot instead of on the first screen someone opens.

05

Publish it, and make it survive a reboot

SERVICESUITE-IN

```
tailscale funnel --bg --https=8443 8787
tailscale funnel status          # must list the mapping

powershell -ExecutionPolicy Bypass -File install-services.ps1 -Sql
```

Only `-Sql`. The CRB relay does not belong on this box. Then reboot and confirm it comes back — that is the only real test.

06

Point Vercel at it

CUTOVER MOMENT

```
SERVICESUITE_RELAY_URL = https://servicesuite-in.tail10c441.ts.net:8443
CRB_RELAY_URL          = https://salesmaster.tail10c441.ts.net # unchanged
```

Then **redploy once** — environment variables are snapshotted into a deployment. After this, restarting a relay never needs a redeploy again.

Note the `:8443`. Dropping it does not fail cleanly; it would reach whatever else answers 443 on that host.

07

Verify from outside, not from the tailnet

WORKSTATION

```
npm run test:relay      # liveness, then a real signed read
npm run test:crb:prod   # a real bureau call, still via salesmaster
npm run test:live       # live / degraded / broken, per subdomain
```

These use the same client code Vercel uses, so a pass means the deployment works rather than that your laptop does.

08

Only now, retire the old relays

AFTER GREEN

Two stale funnels to clear, each on its own box, each named explicitly:

```
# on servicesuite - the relay production used until step 6
tailscale funnel --https=443 off

# on salesmaster - the stale SQL relay on 8443. LEAVE 443 ALONE:
# that is the CRB relay and it is the live bureau link.
tailscale funnel --https=8443 off
tailscale funnel status      # 443 must STILL be listed
```

Stop the superseded Windows services on both boxes so a reboot does not quietly resurrect a second relay holding pools against a lender's database.

09

Apply the tailnet policy

ITS OWN CHANGE WINDOW

Written to `relay-host\tailnet-policy.hjson`. Do it as a separate change, after the cutover is green, so a network lockdown and a relay move are never being debugged at the same time.

GATE The write allowlist, and what it now refuses

The allowlist existed but lived only in `.env`, which has three holes. It is now pinned in `sql-relay.ts` as a reviewed table, and the environment variable is demoted to a per-host switch that can **narrow that list and never widen it**.

The ceiling

WRITE_ALLOWLIST • in source, in git

Every write this proxy is ever capable of. A grant names the orgs it covers, the exact procedure, the only parameters accepted, and why it exists. Raising it requires a code change and a review.

The switch

SQL_RELAY_ALLOW_PROCS • per host

Which pinned grants are live on this box. Unset means none. A name that matches no grant does nothing at all — and the relay says so at boot rather than 403-ing forever in silence.

Both keys must turn. The practical consequence is that **deploying the new file changes no host's posture on its own**, which is what lets it be rolled out without a maintenance window.

Proven against the live relay

Run on this workstation against Micromart's real book. Every case below is a refusal, so nothing was written:

ALLOWLIST BEHAVIOUR · VERIFIED 11 SEP 2026

REQUEST	RELAY'S ANSWER	
proc dbo.sp_ApproveLoan	403	not in this relay's write allowlist
SMS proc, org axe	403	permitted for micromart, micromart-fintech — not for “axe”
SMS proc + extra param	403	accepts only receiver, bodyMessage, companyid — refused unexpected: smsProviderId
exec UPDATE Loans ...	403	arbitrary SQL is never covered by the allowlist, at any setting
read SELECT 1	200	1 row · 82 ms — reads unaffected

THE ORG SCOPE IS THE NEW PART, AND IT MATTERS MOST

The old list admitted a procedure *name* on every org the relay could route to. The same relay reaches Micromart and Axe, so a door opened for one lender's outbox stood open on the other lender's server. Grants are now per-org, and Axe is deliberately absent from the only grant that exists.

CRB Micromart's bureau calls may leave from two boxes only

Your rule is now enforced in `crb-relay.mjs` rather than printed. Until today the egress address was checked at boot, logged, and then ignored — the relay would happily serve bureau traffic from a box Metropolis has never heard of.

- The gate runs **after** signature verification, so an unauthenticated caller learns nothing about the topology, and **before** the outbound call, so an unregistered host fails loudly here instead of silently at the bureau.
- It keys on the **egress IP**, because that is what Metropolis actually checks. A hostname is what we call the box; the address is what the bureau sees, and the two come apart exactly when it matters — an exit node, a dropped TAP adapter, a restored VM.
- It re-reads every five minutes. Fail closed when the address has never been determined; keep serving on a transient lookup failure if a good

reading is under an hour old, so a third party's outage cannot take the bureau link down.

Which host at Micromart's site — measured, not assumed

`lms` is a guest spun up inside the `services` box, so “the Micromart site” is two tailnet nodes sharing one `102.210.148.0/24`. They do not present the same address to the internet. Read from each peer's `CurAddr` on 11 Sep 2026:

EGRESS ADDRESSES · TAILSCALE STATUS --JSON			
NODE	TAILNET	EGRESS	METROPOL 2026-08-27 FORM
salesmaster	100.106.100.118	102.214.69.233	ON IT verified 8/8 ENTITLED
lms	100.92.236.116	102.210.148.110	ON IT the registered host at this site
services	100.72.35.56	102.210.148.124	NOT ON IT

THIS RESOLVES THE NEAR-MISS THAT HAS BEEN SITTING IN THE CODE

`.110` versus `.124` was never a wrong digit on a form. They are two different hosts at one site, and `lms` is the one Metropol registered. So your rule — bureau calls leave from `salesmaster` or from Micromart's own site — is satisfiable today, using `lms` as the second origin rather than `services`. Both are now in the relay's permitted set; `services` is marked unverified and says so at boot.

Nothing needs to move: `salesmaster` is serving the bureau and is verified. `lms` becomes the standby that would actually work if you ever needed it — which is what it was not, an hour ago.

If you do want `services` itself entitled, that is a bureau-side task: ask Metropol to add `102.210.148.124`. Settle whether it is already open without guessing — port 22225 does not open for unregistered addresses at all, so a completed handshake is proof of entitlement and a hang is proof of the opposite:

```
Test-NetConnection api.metropol.co.ke -Port 22225
```

NEVER SET AN EXIT NODE ON A BUREAU HOST

Tailscale does not change your egress by default, but an exit node would — and the egress address is the entire reason these two boxes are special. It would break the whitelist silently, which is the failure mode this whole section exists to prevent. The relay now re-reads its own address every five minutes and refuses when it changes, so this fails loudly instead. Do not rely on that as a licence to try it.

ACL Managing the tailnet itself

Eight nodes currently run the default policy — every node can reach every port on every other node. That means a workstation, a spare VPS and two IIS boxes all hold a standing route to two lenders' live SQL Servers, and the relay's careful allowlist is bypassed by anyone who can open a TDS socket directly.

relay-host\tailnet-policy.hujson is the other half of that control. Paste it into the tailnet's policy editor. The shape:

TAGS AND WHAT THEY ARE PERMITTED		
TAG	NODES	GRANT
tag:lender-db	services, axemicro	Receives TDS from tag:relay and nothing else. Originates nothing.
tag:relay	servicesuite-in, servicesuite	The only route to a lender's book. May run Funnel.
tag:bureau	salesmaster, lms, services	May run Funnel. Carries the CRB relay. lms is here on the egress evidence above, not as a spare.
tag:admin-ws	pc	RDP and web to the estate. Break-glass TDS, marked for deletion.
tag:spare	vmi3298281	Nothing. Asserted by a test.

- **Funnel becomes a permission, not a habit.** Five of eight nodes have it enabled today, three for nothing. The funnel node attribute means a node without it cannot turn Funnel on whatever is typed on the box.
- **The policy tests itself.** The tests block runs on Save and refuses the policy if any assertion fails — including “one lender's server cannot reach the other's”, which is a promise the Interchange makes to its members and should be true at the packet level, not only in a diagram.
- **Tagging a device stops its key expiring,** which is separately what you want on a server.

ORDER OF OPERATIONS, OR YOU WILL LOCK YOURSELF OUT

A device with no tag matches none of the grants. Save the file first — nothing changes while nothing is tagged. Then tag devices one at a time, **workstation first**, so you keep a way in. Watch Admin → Logs for denied flows for a day before calling it done. Reverting is one Save away; keep the default policy in a scratch file.

NEXT What this unblocks, and what is still yours to decide

With a relay on `servicesuite-in` the Interchange node has a home: the ingest that reads a member's book, derives a `subject_token` through the blinded OPRF and publishes tokens plus aggregates runs on the same box, over the same tailnet route, with the same credentials. No inbound port on either lender's machine — the node dials out to SQL and pushes to the Registry over HTTPS.

The foundation the Axe ↔ Micromart exchange still needs

- A read-only login per server. The suite connects today as `SystemUser1` on Axe and `service_suite_user` on Micromart — application logins that can write and can read every entity. Ten unrelated companies share the `.198` book separated by nothing but `EntityId`, so scope is a security boundary, not a filter. Create `interchange_node`, give it a view per entity with the `EntityId` baked inside the view, `GRANT` the schema and `DENY` everything else.
- Sprint 3's done-when is “on the real Micromart book”. The engine, the Bloom screening and the CRB 2.0 envelopes are all built and tested; the 400 ms was measured against four hundred invented borrowers. The relay is what points it at `Serviceconnect`.
- Both Axe entities stay **SHADOW** until their first ingest commits. Contributing, not yet querying — the reciprocity rule enforced in the policy engine rather than in a contract.

Decisions only you can make

- Member codes. `EntityId` 3003 is “Micromart Check off” on one server and “Axe — Boresha” on the other. A bare `KE/LENDER/3003` names two different lenders and whichever registered first silently owns the other's exposure. Key on the KRA PIN in `BsEntity.EntityTaxRef` instead, and migrate the fifteen existing members now while the message log is short. This gets harder every week it waits.
- Whether to keep the break-glass TDS grant from the workstation once the relay is serving. It is how live verification is done from this repo today; it is also precisely the hole the policy closes.
- Whether `services` needs entitling at all. `lms` already covers Micromart's site with a registered address. Adding `102.210.148.124` to Metropolis's form is only worth doing if you specifically want the relay on `services` rather than on the guest inside it.
- Postgres on `213.148.17.228:2020`. Whether it answers from the public internet, and if so, moving it behind the tailnet. Same argument as the SQL relay, and a shorter one.

ONE THING I DID NOT TOUCH

Nothing here was applied to any server. The code changes are in the working tree on this workstation and are unstaged; the relays on `servicesuite` and `salesmaster` are running the builds they were running this morning. Every step above is yours to execute on the box you are logged into.